MATH-1014 T10B

Tutorial 11: Advanced Convergence Tests and Strategies

WANG Yixiao

Department of Mathematics

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Today's Roadmap

1. Concept Review: The Essence of Convergence Tests
2. Practice Problems

Review: The Anatomy of a Series

Do not blindly guess which test to use. The mathematical structure (the "anatomy") of the general term a_n usually dictates the exact tool you need. Think of tests as specific "measuring tools".

Visual Cues for Choosing a Convergence Test

- **Rational/Algebraic Functions:** $\frac{\text{Poly}}{\text{Poly}}$ or roots. Use the **Limit Comparison Test (LCT)**. p-series are your "rulers".
- **Factorials ($n!$) or Exponentials (c^n):** Rapid growth. The **Ratio Test** is the "exponential scanner".
- **Nested n-th Powers:** Expressions like $(f(n))^n$. Use the **Root Test** to instantly neutralize the outer exponent.
- **Alternating Signs:** $(-1)^n$. Trigger the **Alternating Series Test (AST)**.

Review: The Two Tiers of Convergence

Not all convergent series are created equal. We must distinguish between "strong" and "fragile" convergence.

Absolute Convergence (The Gold Standard)

A series $\sum a_n$ is **absolutely convergent** if $\sum |a_n|$ converges.

- **Essence:** The series converges purely because its terms shrink fast enough, regardless of their signs.

Conditional Convergence (The Fragile State)

A series is **conditionally convergent** if it converges, but $\sum |a_n|$ diverges.

- **Essence:** The magnitudes $|a_n|$ are too large to converge on their own. The series *only* survives because the alternating positive and negative terms cancel each other out.

Problem 1: Absolute vs. Conditional Convergence

Problem 1.
Determine whether the following series is absolutely convergent, conditionally convergent, or divergent:

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2+4}$$

Strategy: Always test for Absolute Convergence first. If that fails, test for Conditional Convergence using the Alternating Series Test.

Solution 1: Checking Absolute Convergence

Step 1: Test the series of absolute values.
We examine $\sum_{n=1}^{\infty} \left|(-1)^{n-1}\frac{n}{n^2+4}\right| = \sum_{n=1}^{\infty} \frac{n}{n^2+4}$.

Notice that for large n, $\frac{n}{n^2+4}$ behaves like $\frac{n}{n^2} = \frac{1}{n}$. We use the **Limit Comparison Test** with the divergent Harmonic Series $\sum \frac{1}{n}$:

$$\lim_{n\to\infty} \frac{\frac{n}{n^2+4}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n^2}{n^2+4} = 1 > 0$$

Since $\sum \frac{1}{n}$ diverges, our absolute series also diverges.
Conclusion 1: The series is **NOT** absolutely convergent.

Solution 1: Checking Conditional Convergence

Step 2: Apply the Alternating Series Test (AST).
Let $b_n = \frac{n}{n^2+4}$. We verify two conditions:

1. **Decreasing:** Let $f(x) = \frac{x}{x^2+4}$. Taking the derivative:

$$f'(x) = \frac{(x^2+4) - x(2x)}{(x^2+4)^2} = \frac{4 - x^2}{(x^2+4)^2}$$

   For $x > 2$, $f'(x) < 0$. Sequence is eventually decreasing.

2. **Limit is zero:** $\lim_{n\to\infty} \frac{n}{n^2+4} = 0$.

Final Conclusion: By the AST, the series converges. Since it converges but not absolutely, it is **conditionally convergent**.

Problem 2: The Ratio Test

Problem 2.
Test the following series for absolute convergence:

$$\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{3^n}$$

Hint: The presence of exponentials 3^n strongly suggests the Ratio Test.

Solution 2: Applying Ratio Test

Step 1: Set up the ratio limit.

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{(n+1)^3}{3^{n+1}}}{\frac{n^3}{3^n}}\right|$$

Step 2: Simplify and Evaluate.

$$= \lim_{n\to\infty}\left(\frac{3^n}{3^{n+1}}\cdot\frac{(n+1)^3}{n^3}\right) = \lim_{n\to\infty}\frac{1}{3}\cdot\left(1+\frac{1}{n}\right)^3$$

Since $\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^3 = 1$, the overall limit is:

$$L = \frac{1}{3} < 1$$

Conclusion: By the Ratio Test, the series is **absolutely convergent**.

Problem 3: The Root Test

Problem 3.

Test the convergence of the series:

$$\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2} \right)^n$$

Solution 3: Applying Root Test

Step 1: Set up the root limit.
We calculate $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$.

$$L = \lim_{n\to\infty} \sqrt[n]{\left(\frac{2n+3}{3n+2}\right)^n} = \lim_{n\to\infty} \frac{2n+3}{3n+2}$$

Step 2: Evaluate. Divide the numerator and denominator by n:

$$L = \lim_{n\to\infty} \frac{2+\frac{3}{n}}{3+\frac{2}{n}} = \frac{2}{3}$$

Conclusion: Since $L = \frac{2}{3} < 1$, the given series **converges** by the Root Test.

Problem 4: Limit Comparison with Roots

Problem 4.
Determine whether the following series converges or diverges:

$$\sum_{n=1}^{\infty} \frac{\sqrt{n^3+1}}{3n^3+4n^2+2}$$

Solution 4: Applying LCT

Step 1: Identify the dominant terms.
For large n, numerator $\approx n^{3/2}$, denominator $\approx 3n^3$.
Comparison series: $b_n = \frac{n^{3/2}}{3n^3} = \frac{1}{3n^{3/2}}$.
Step 2: Limit Comparison Test.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{\sqrt{n^3+1}}{3n^3+4n^2+2}}{\frac{1}{3n^{3/2}}} = \lim_{n\to\infty} \frac{3\sqrt{n^6+n^3}}{3n^3+4n^2+2} = 1 > 0$$

Conclusion: Both share the same fate. Since $\sum \frac{1}{3n^{3/2}}$ is a convergent p-series ($p = 1.5 > 1$), our original series **converges**.

Problem 5: Advanced Integral Test

Problem 5.
Determine whether the following series converges or diverges:

$$\sum_{n=3}^{\infty} \frac{1}{n\ln(n)\sqrt{\ln(\ln(n))}}$$

Observation: The terms are positive, continuous, and decreasing. When you see nested logarithms accompanied by $1/n$, it perfectly sets up the chain rule for the **Integral Test**.

Solution 5: Nested Integration

Step 1: Set up the improper integral.

$$\int_3^\infty \frac{1}{x\ln(x)\sqrt{\ln(\ln(x))}}dx$$

Step 2: U-Substitution. Let $u = \ln(\ln(x))$. Then by the chain rule:

$$du = \frac{1}{\ln(x)} \cdot \frac{1}{x}dx$$

Step 3: Evaluate.

$$\int \frac{1}{\sqrt{u}}du = \int u^{-1/2}du = 2u^{1/2} = 2\sqrt{\ln(\ln(x))}$$

Evaluating from 3 to $t \to \infty$:

$$\lim_{t\to\infty}\left[2\sqrt{\ln(\ln(t))} - 2\sqrt{\ln(\ln(3))}\right] = \infty$$

Conclusion: The integral diverges, so the series **diverges**.

Problem 6: Rationalization & Alternating Series

Problem 6.
Test for absolute and conditional convergence:

$$\sum_{n=1}^{\infty} (-1)^n \left(\sqrt{n+1} - \sqrt{n}\right)$$

Hint: $\infty - \infty$ is indeterminate. You cannot determine the behavior of a_n without rationalizing the expression first.

Solution 6: Algebraic Manipulation

Step 1: Rationalize the term.

$$a_n = (\sqrt{n+1} - \sqrt{n}) \cdot \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

The series is $\sum (-1)^n \frac{1}{\sqrt{n+1}+\sqrt{n}}$.

Step 2: Check Absolute Convergence. $\sum |a_n| = \sum \frac{1}{\sqrt{n+1}+\sqrt{n}}$. For large n, $|a_n| \approx \frac{1}{2\sqrt{n}}$. Using LCT with $b_n = \frac{1}{\sqrt{n}}$ ($p = 1/2 < 1$, diverges), we find the absolute series **diverges**.

Step 3: Check Conditional Convergence (AST). Let $b_n = \frac{1}{\sqrt{n+1}+\sqrt{n}}$. 1. Decreasing? Yes, denominator increases as n increases. 2. Limit 0? Yes, $\lim_{n\to\infty} \frac{1}{\infty} = 0$. By AST, the series converges. Thus, it is **conditionally convergent**.

Problem 7: Riemann Sum Limit (Series to Integral)

Problem 7.
Evaluate the limit of the sequence of sums:

$$\lim_{n\to\infty} \sum_{i=1}^{n} \frac{i}{n^2 + i^2}$$

Strategic Shift

This is not a geometric series, and Squeeze Theorem fails. An infinite sum limit of this specific rational structure is the definition of a **Definite Integral (Riemann Sum)**.

Solution 7: Constructing the Integral

Step 1: Factor out $1/n$ to expose Δx. We want to express it as $\sum f(x_i)\Delta x$. Divide numerator and denominator by n^2:

$$\lim_{n\to\infty}\sum_{i=1}^{n}\frac{\frac{i}{n^2}}{1+\left(\frac{i}{n}\right)^2}=\lim_{n\to\infty}\sum_{i=1}^{n}\frac{\frac{i}{n}}{1+\left(\frac{i}{n}\right)^2}\cdot\frac{1}{n}$$

Step 2: Map to the Definite Integral. Let $\Delta x=\frac{1}{n}$ and $x_i=\frac{i}{n}$ on $[0,1]$. Then $f(x)=\frac{x}{1+x^2}$.

$$\lim_{n\to\infty}\cdots=\int_0^1\frac{x}{1+x^2}dx$$

Step 3: Evaluate. Let $u=1+x^2, du=2xdx\implies\frac{1}{2}\int\frac{1}{u}du$.

$$=\frac{1}{2}\left[\ln(1+x^2)\right]_0^1=\mathbf{\frac{1}{2}}\ln(\mathbf{2})$$

Problem 8: Improper Integral (Double Singularity)

Problem 8.
Determine whether the following improper integral converges or diverges:

$$\int_0^\infty \frac{e^{-x}}{\sqrt{x}} dx$$

Warning: Hidden Asymptote

This integral is "improper" for **two** reasons: the upper bound is ∞, AND the integrand has a vertical asymptote at $x = 0$. You must split the integral into two parts and test each separately!

Solution 8: Splitting and Direct Comparison

Step 1: Split the integral at any intermediate point, say $x = 1$.

$$\int_0^{\infty} = \int_0^1 \frac{e^{-x}}{\sqrt{x}} dx + \int_1^{\infty} \frac{e^{-x}}{\sqrt{x}} dx$$

Step 2: Test Part 1 (The Singularity at $x = 0$). On the interval $(0, 1]$, $e^{-x} \leq 1$. Therefore, $0 \leq \frac{e^{-x}}{\sqrt{x}} \leq \frac{1}{\sqrt{x}}$. Since $\int_0^1 \frac{1}{x^{1/2}} dx$ is a convergent p-integral $(p = 1/2 < 1)$, the first part **converges** by the Direct Comparison Test.

Step 3: Test Part 2 (The Tail as $x \to \infty$). On the interval $[1, \infty)$, $\sqrt{x} \geq 1 \implies \frac{1}{\sqrt{x}} \leq 1$. Therefore, $0 \leq \frac{e^{-x}}{\sqrt{x}} \leq e^{-x}$. Since $\int_1^{\infty} e^{-x} dx$ converges, the second part **converges** by the Direct Comparison Test.

Conclusion: Because both independent parts converge, the entire integral **converges**.

Problem 9: The Ultimate Root Test

Problem 9.
Determine whether the following series converges or diverges:

$$\sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^{n^2}$$

Hint: A function of n raised to an n-based power is the classic signal to deploy the Root Test. Watch out for the limit definition of e!

Solution 9: Applying the Root Test

Step 1: Apply the n-th root to the general term.

$$L = \lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \left(\left(\frac{n}{n+1}\right)^{n^2}\right)^{1/n}$$

Step 2: Simplify the exponent.

$$L = \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^{n}$$

Step 3: Algebraic manipulation to reveal e. Rewrite the fraction to match the standard limit definition $\lim_{n\to\infty}(1+\frac{x}{n})^n = e^x$:

$$L = \lim_{n\to\infty} \left(\frac{n+1-1}{n+1}\right)^{n} = \lim_{n\to\infty} \left(1 - \frac{1}{n+1}\right)^{n}$$

$$L = \lim_{n\to\infty} \left(1 - \frac{1}{n+1}\right)^{n+1} \cdot \left(1 - \frac{1}{n+1}\right)^{-1}$$

$$L = e^{-1} \cdot 1 = \frac{1}{e}$$

Problem 10: Improper Integral via Parts

Problem 10.
Evaluate the following improper integral exactly, or show that it diverges:

$$\int_1^\infty \frac{\ln x}{x^2}\,dx$$

Strategy: You cannot compare this easily to get an exact value. You must evaluate it using Integration by Parts, then carefully apply limits using L'Hôpital's Rule.

Solution 10: Integration by Parts & L'Hôpital's

Step 1: Integration by Parts for the indefinite integral. Let $u = \ln x$ and $dv = x^{-2}dx$. Then $du = \frac{1}{x}dx$ and $v = -x^{-1}$.

$$\int \frac{\ln x}{x^2}dx = -\frac{\ln x}{x} - \int -x^{-1} \cdot \frac{1}{x}dx = -\frac{\ln x}{x} - \frac{1}{x}$$

Step 2: Apply the limits of integration ($t \to \infty$).

$$\int_1^\infty \frac{\ln x}{x^2}dx = \lim_{t\to\infty} \left[-\frac{\ln x}{x} - \frac{1}{x}\right]_1^t$$

$$= \lim_{t\to\infty} \left(-\frac{\ln t}{t} - \frac{1}{t}\right) - \left(-\frac{\ln 1}{1} - \frac{1}{1}\right)$$

Step 3: Evaluate the limit. The term $\lim_{t\to\infty} \frac{\ln t}{t}$ is of form $\frac{\infty}{\infty}$. By L'Hôpital's Rule:

$$\lim_{t\to\infty} \frac{1/t}{1} = 0$$

So the entire limit evaluates to $(0-0)-(0-1) = \mathbf{1}$. The integral **converges to 1**.

Problem 11: The Hidden Telescoping Series

Problem 11.
Find the exact sum of the series:

$$\sum_{n=2}^{\infty} \ln\left(1 - \frac{1}{n^2}\right)$$

Hint

Any time you are asked to find the **exact sum** of a series that isn't geometric or a known Maclaurin series, you are almost certainly looking at a **Telescoping Series**. Use logarithm properties to break the term apart!

Solution 11: Factorization and Expansion

Step 1: Manipulate the argument inside the logarithm.

$$\ln\left(\frac{n^2-1}{n^2}\right) = \ln\left(\frac{(n-1)(n+1)}{n \cdot n}\right)$$

Using log properties $\ln(AB/CD) = \ln A + \ln B - \ln C - \ln D$:

$$a_n = \ln(n-1) - \ln(n) + \ln(n+1) - \ln(n)$$

Step 2: Write out the partial sum S_N to find the cancellation pattern.

$$\begin{aligned}
n = 2: &\quad \ln(1) - \ln(2) + \ln(3) - \ln(2) \\
n = 3: &\quad \ln(2) - \ln(3) + \ln(4) - \ln(3) \\
n = 4: &\quad \ln(3) - \ln(4) + \ln(5) - \ln(4) \\
\ldots & \\
n = N: &\quad \ln(N-1) - \ln(N) + \ln(N+1) - \ln(N)
\end{aligned}$$

Solution 11: The Final Limit

Step 3: Cancel terms vertically. Notice how the positive terms in one row cancel perfectly with the negative terms in the surrounding rows. Everything collapses except the very first non-zero terms and the very last terms:

$$S_N = \ln(1) - \ln(2) + \ln(N+1) - \ln(N)$$

Since $\ln(1) = 0$, we have:

$$S_N = -\ln(2) + \ln\left(\frac{N+1}{N}\right)$$

Step 4: Take the limit as $N \to \infty$.

$$\lim_{N\to\infty} S_N = -\ln(2) + \lim_{N\to\infty} \ln\left(1 + \frac{1}{N}\right)$$

$$= -\ln(2) + \ln(1) = -\ln(\mathbf{2})$$

The series converges exactly to $-\ln(2)$.

Problem 12: Taylor Expansion & LCT (Advanced)

Problem 12.
Determine whether the following series converges or diverges:

$$\sum_{n=1}^{\infty}\left(\frac{1}{n}-\sin\left(\frac{1}{n}\right)\right)$$

Why this is tricky

As $n \to \infty$, $a_n \to 0 - 0 = 0$. The test for divergence fails. But how fast does it go to zero? We need **Maclaurin series** to reveal the hidden order of magnitude.

Solution 12: Revealing the Order of Growth

Step 1: Use the Maclaurin series for $\sin(x)$. We know $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$
Let $x = \frac{1}{n}$. For large n (small x), we have:

$$\sin\left(\frac{1}{n}\right) \approx \frac{1}{n} - \frac{1}{6n^3}$$

Step 2: Substitute back into a_n.

$$a_n = \frac{1}{n} - \left(\frac{1}{n} - \frac{1}{6n^3} + \text{higher order terms}\right) \approx \frac{1}{6n^3}$$

Step 3: Limit Comparison Test. Let $b_n = \frac{1}{n^3}$.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{n} - \sin(1/n)}{1/n^3} \overset{\text{L'H}}{=} \dots = \frac{1}{6} > 0$$

Since $\sum \frac{1}{n^3}$ converges ($p = 3 > 1$), the series **converges**.

Problem 13: The "Boss-Level" Exam Classic

Problem 13.
Determine whether the following series is absolutely convergent, conditionally convergent, or divergent:

$$\sum_{n=1}^{\infty}(-1)^n\left[e-\left(1+\frac{1}{n}\right)^n\right]$$

The Ultimate Trap

We know $\lim_{n\to\infty}(1+1/n)^n = e$, so the general term goes to zero. But how fast? To use the Limit Comparison Test for absolute convergence, you must find the hidden p-value using nested Taylor expansions!

Solution 13: Absolute Convergence (Taylor Analysis)

Step 1: Find the asymptotic behavior of $b_n = e - (1 + 1/n)^n$.
First, rewrite the exponential term using logarithms: $(1 + 1/n)^n = e^{n \ln(1+1/n)}$.

Use the Taylor series $\ln(1 + x) \approx x - \frac{x^2}{2}$ (for small $x = 1/n$):

$$n \ln\left(1 + \frac{1}{n}\right) \approx n\left(\frac{1}{n} - \frac{1}{2n^2}\right) = 1 - \frac{1}{2n}$$

Substitute this back into the base e expression:

$$(1 + 1/n)^n \approx e^{1-\frac{1}{2n}} = e \cdot e^{-\frac{1}{2n}}$$

Now use the Taylor series $e^x \approx 1 + x$ (for small $x = -1/(2n)$):

$$e \cdot e^{-\frac{1}{2n}} \approx e\left(1 - \frac{1}{2n}\right) = e - \frac{e}{2n}$$

Solution 13: The Final Verdict

Step 2: Execute the Limit Comparison Test (LCT).
From Step 1, $b_n = e - \left(1 + \frac{1}{n}\right)^n \approx e - \left(e - \frac{e}{2n}\right) = \frac{e}{2n}$.
This means the absolute series $\sum b_n$ behaves exactly like the harmonic series $\sum \frac{1}{n}$. By LCT with $\sum \frac{1}{n}$, the absolute series **diverges**.

Step 3: Check Conditional Convergence (AST).

- We know the sequence $(1 + 1/n)^n$ is strictly increasing towards e.
- Therefore, $b_n = e - (1 + 1/n)^n$ is strictly **decreasing** and always **positive**.
- We already established $\lim_{n\to\infty} b_n = 0$.

By the Alternating Series Test, the series converges.
Final Verdict: Conditionally Convergent.

THANK YOU!